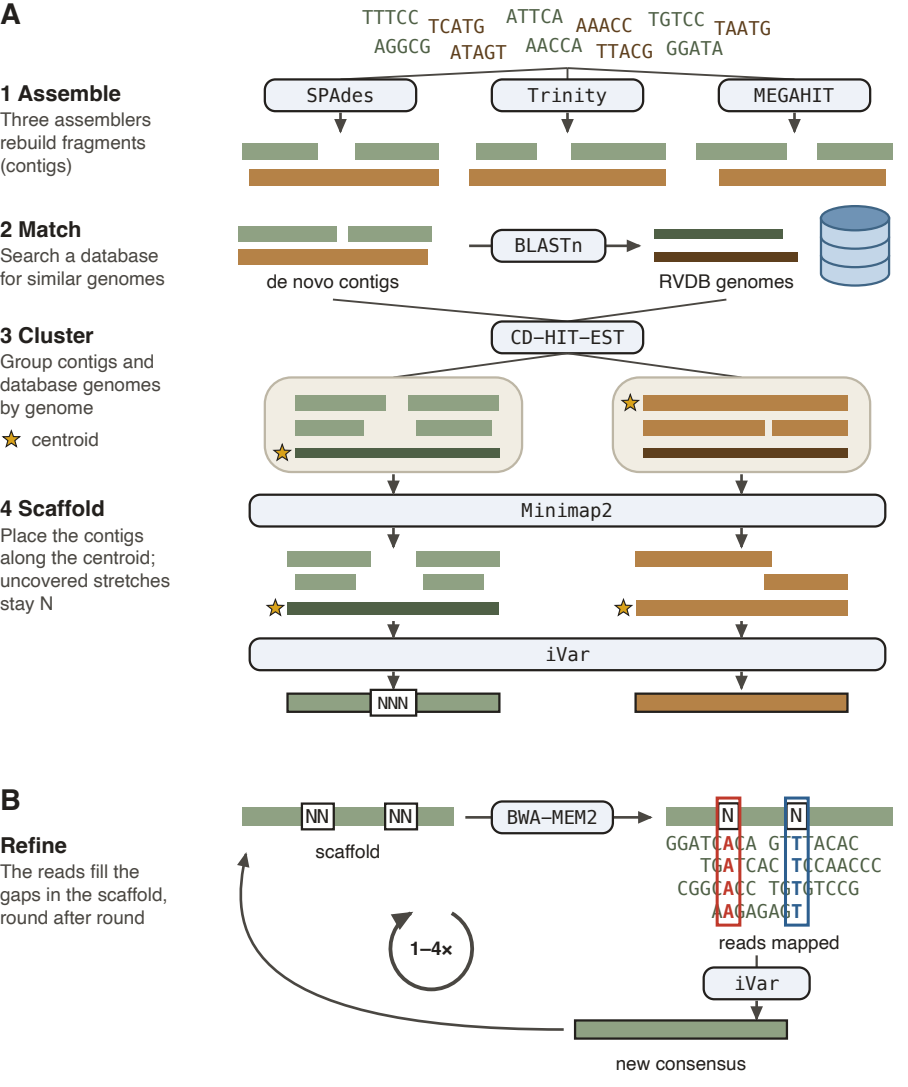
A
TTTCC TCATG ATTCA AAACC TGTCC TAATG
AGGCG ATAGT AACCA TTACG GGATA
1 Assemble
Three assemblers rebuild fragments (contigs)
SPAdes
Trinity
MEGAHIT
2 Match
Search a database for similar genomes
de novo contigs
BLASTn
RVDB genomes
3 Cluster
Group contigs and database genomes by genome
centroid
CD-HIT-EST
4 Scaffold
Place the contigs along the centroid; uncovered stretches stay N
Minimap2
iVar
NNN
B
Refine
The reads fill the gaps in the scaffold, round after round
NN
NN
scaffold
BWA-MEM2
N
N
GGATCACA GTTTACAC
TGATCAC TCCAACCC
CGGCACC TGTGTCCG
AAGAGAGT
reads mapped
1–4×
iVar
new consensus